Waseem Raza, PhD

Gen AI Engineer | Agentic AI, RAG & Production ML

Enterprise GenAI | Data Science | LLM Applications | Scalable AI Systems

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LinkedIn | GitHub | Google Scholar | Portfolio

Professional Summary

Generative AI and Machine Learning Engineer with a PhD in Electrical and Computer Engineering and 7+ years of combined industry and research experience building data-driven systems from data preparation through model validation and operational integration. Hands-on experience spans enterprise LLM applications, retrieval-augmented generation (RAG), vector retrieval, agentic workflows, prompt engineering, multimodal/document AI, Python data pipelines, cloud analytics, and production-oriented ML. Current work at AT&T includes end-to-end AI diagnostics for critical E911 service assurance, multimodal GenAI prototypes, cross-domain model validation, and stakeholder-driven delivery. Strong background in LangChain, LangGraph, LlamaIndex, OpenAI APIs, AWS Bedrock, Azure AI services, Docker, Kubernetes, GitHub Actions, CI/CD, model evaluation, and reliable ML under distribution shift.

Technical Skills

Enterprise GenAI & RAG: LLM applications, RAG, document ingestion, chunking, embeddings, semantic search, vector databases, ChromaDB, FAISS, HNSW, grounded generation, prompt engineering, structured outputs.
Agentic AI & Orchestration: LangChain, LangGraph, LlamaIndex, tool calling, state and memory, conditional routing, ReAct-style reasoning, CrewAI, AutoGen/AG2, BeeAI, multi-agent coordination, Model Context Protocol (MCP).
LLM Platforms & Models: OpenAI APIs, AWS Bedrock, Azure OpenAI, Microsoft Foundry, Hugging Face Transformers, foundation models, transformer architectures, fine-tuning and model-adaptation concepts.
LLMOps, Evaluation & Reliability: LLM/model evaluation, retrieval evaluation, error analysis, cross-domain validation, failure-mode assessment, responsible AI, model safety concepts, performance and reliability testing, reproducible experimentation.
APIs, MLOps & Software Engineering: Python, REST APIs, Flask, Docker, Kubernetes, OpenShift, Git, GitHub Actions, CI/CD, Linux, testing, system design, distributed-system and microservice concepts.
Data Science & Engineering: SQL, Pandas, NumPy, scikit-learn, PyTorch, TensorFlow, XGBoost, Azure Databricks, Apache Spark/PySpark, Synapse Analytics, Azure Functions, ETL/ELT, data validation, feature engineering, forecasting and predictive modeling.
Multimodal & Document AI: OCR, document intelligence, named-entity and table extraction, OpenAI Whisper, speech-to-text, computer vision, structured information extraction, multimodal workflow integration.

Professional Experience

Machine Learning Engineer **02/2025 – Present**
AT&T, contractor via Innovatix Inc. *Bedminster, NJ*

- Own end-to-end development, testing, and operational integration of an AI-driven diagnostic system for critical E911 service-assurance workflows, improving automated root-cause classification accuracy from 20% to 80%.
- Develop multimodal GenAI-assisted diagnostic prototypes that combine work-order text, location information, network context, and operational signals for VoNR and 5G service-assurance use cases.
- Design domain-adaptive neural models and reusable data-preparation workflows that improve generalization across heterogeneous operational domains and distribution shifts.
- Build model-validation, error-analysis, and failure-mode assessment procedures to compare performance across domains and support reliable production deployment.
- Partner with network engineers, software teams, and public-safety stakeholders to translate operational requirements into measurable AI objectives, validated data assets, and deployable workflows.

Data Scientist and AI/ML Engineer **09/2024 – 12/2024**
Impac Exploration Services *Weatherford, OK*

- Built an LLM-enabled analytical pipeline for extracting and interpreting information from geochemical well data using Python and Azure-based data services.
- Developed scalable ingestion, cleansing, transformation, and validation workflows using Azure Databricks, Azure Functions, and Synapse Analytics.
- Automated ETL processing for large exploration datasets, reducing data-preparation time by 40% and improving reproducibility; implemented validation checks that improved reliability of analytical outputs by 20%.
- Collaborated with geologists to translate domain requirements into AI-ready datasets and structured analytical outputs.

Additional AI/ML Experience

Graduate Research Assistant **09/2019 – 07/2025**
AI4Networks, University of Oklahoma *Tulsa, OK*

- Designed reusable Python simulation and data-generation platforms for large-scale experimentation, including 3GPP-compliant network simulation and an AI-enabled 5G testbed.
- Built transfer-learning, adversarial-learning, GAN, autoencoder, and reinforcement-learning pipelines for sparse-data and cross-domain modeling; led 100+ experiments from formulation through validation.
- Developed robustness-evaluation frameworks for distribution shift, missing data, and limited labeled samples, and published peer-reviewed research on reliable AI-enabled systems.
- Mentored junior researchers in ML implementation, experimental design, reproducible analysis, and technical writing.

Advanced-Degrees Intern **06/2024 – 08/2024**
AT&T Labs *Austin, TX*

- Analyzed 500K+ LTE/5G-NR diagnostic logs and applied ML methods to detect anomalies, reducing diagnostic time by 20%.
- Conducted structured analysis of large operational datasets and communicated technical findings to engineering teams supporting next-generation wireless systems.

Selected GenAI Projects

Agentic AI and RAG Applications *Portfolio Project | 2026*

Built grounded GenAI workflows that combine document retrieval, vector search, LangGraph-style state and routing, tool-using agents, memory, evaluation, and multi-agent handoffs. The project emphasizes decomposition of multi-step tasks, controlled execution, traceable context flow, and reliable responses rather than single unconstrained model calls.

Tools & skills: Python, LangGraph, LangChain, RAG, vector databases, embeddings, semantic retrieval, tool calling, agent memory, evaluation, multi-agent workflow design.

Multi-Agent Orchestration with LangGraph, CrewAI & AutoGen *Certification Project | 2026*

Implemented stateful agent workflows with tool calling, memory, conditional routing, ReAct-style reasoning, specialized-agent handoffs, and synthesis across LangGraph, CrewAI, AutoGen/AG2, and BeeAI. Focused on orchestration patterns that make multi-agent behavior explicit and reusable.

Tools & skills: LangGraph, CrewAI, AutoGen/AG2, BeeAI, tool calling, state/memory, conditional routing, ReAct, multi-agent coordination.

RAG Retrieval Stack with ChromaDB, FAISS & LlamaIndex *Certification Project | 2026*

Built retrieval workflows covering document processing, chunking, embeddings, vector indexing, similarity search, advanced retrievers, and interactive RAG application flows. Compared retrieval components and connected retrieved context to LLM generation through LangChain and LlamaIndex orchestration.

Tools & skills: RAG, ChromaDB, FAISS, LlamaIndex, LangChain, embeddings, HNSW, semantic search, retriever design, Gradio.

Amazon Bedrock Generative AI & Agent Workflows *Certification Project | 2026*

Implemented foundation-model prompting, knowledge-base retrieval, RAG integration, model-customization exercises, workflow automation, and intelligent-agent patterns on Amazon Bedrock. Explored cloud-hosted GenAI application patterns with managed model access and retrieval-backed grounding.

Tools & skills: Amazon Bedrock, foundation models, RAG, knowledge bases, LangChain, AI agents, prompt engineering, AWS GenAI services.

AI-Ready Lakehouse Pipeline *Portfolio Project | 2024*

Designed a cloud data pipeline that prepares structured and semi-structured data for analytics and downstream machine-learning workloads, with emphasis on repeatable ingestion, transformation, validation, and scalable processing. The workflow demonstrates the data-readiness layer required before predictive modeling or enterprise GenAI applications can be deployed reliably.

Tools & skills: Azure Databricks, Apache Spark, PySpark, ETL/ELT, schema transformation, data validation, feature preparation, AI-ready datasets, cloud data engineering.

OCR & Document Intelligence Extraction Pipeline *Guided Project | 2025*

Implemented a document-intelligence workflow that converts reports, invoices, and contract-style documents into structured outputs through OCR, layout/table parsing, entity and field extraction, normalization, and validation. Designed the pipeline for downstream analytics and integration rather than isolated text extraction.

Tools & skills: OCR, document intelligence, LLMs, named entity recognition, table extraction, structured JSON/tabular data, normalization, validation, data integration.

Automated Audio Meeting Minutes with Whisper & GPT *Guided Project | 2025*

Built a multimodal workflow that converts recorded audio to text, cleans and chunks transcripts, generates structured summaries, and extracts decisions, owners, and action items. The workflow demonstrates speech-to-text plus LLM-based structured generation for repeatable knowledge-capture tasks.

Tools & skills: Python, OpenAI Whisper, speech-to-text, GPT/LLMs, prompt design, structured outputs, summarization, workflow automation.

System Design Case-Study Portfolio *Guided Project | 2026*

Documented architecture studies that translate product requirements and traffic assumptions into service boundaries, APIs, data models, storage choices, caching strategies, consistency decisions, and reliability/capacity tradeoffs. Provides a software-engineering foundation for deploying AI capabilities as scalable services.

Tools & skills: System design, APIs, distributed systems, scalability, caching, data modeling, reliability, capacity planning, architecture tradeoffs.

Selected Certifications & Professional Development

Generative AI Applications with Amazon Bedrock *Completed 08/2026*

Focus: developed practical understanding of foundation-model access, prompt-driven application logic, managed AWS GenAI services, and cloud patterns for enterprise LLM applications. Reinforced how Bedrock knowledge and application components can support grounded, scalable GenAI workflows. *Skills:* Amazon Bedrock, foundation models, prompt engineering, RAG/application workflows, managed GenAI services.

Agentic AI with LangChain and LangGraph *Completed 02/2026*

Focus: covered stateful agent design, graph orchestration, tool use, memory, routing, and multi-step workflows. Emphasized controllable execution patterns for agents that must maintain context, invoke tools, and coordinate multi-stage tasks. *Skills:* LangChain, LangGraph, tool calling, state/memory, routing, agent orchestration.

Advanced RAG with Vector Databases and Retrievers *Completed 01/2026*
Focus: covered vector databases, indexing, semantic search, advanced retriever design, and retrieval evaluation for grounded LLM applications. Focused on improving context quality before generation by selecting appropriate chunking, indexing, and retrieval strategies. *Skills:* RAG, embeddings, vector search, chunking/indexing, retriever design, retrieval evaluation.

Build AI Agents using MCP *Completed 06/2026*
Focus: connecting language models to external tools, services, and structured context through Model Context Protocol. *Skills:* MCP, tool-using agents, API/tool interfaces, context integration, orchestration.

Generative AI Engineering and Fine-Tuning Transformers *Completed 08/2026*
Focus: pretrained transformers, task adaptation, fine-tuning strategies, evaluation, and training/inference tradeoffs. *Skills:* transformers, foundation models, fine-tuning, transfer learning, model adaptation, LLM evaluation.

Machine Learning *Completed 03/2020*
Focus: supervised and unsupervised learning, regression, classification, regularization, model evaluation, clustering, and practical model-selection considerations. *Skills:* predictive modeling, classification/regression, regularization, clustering, cross-validation concepts, bias-variance analysis.

Education

PhD, Electrical and Computer Engineering *09/2019 – 07/2025*
University of Oklahoma, Norman, Oklahoma, USA

MSc, Telecommunication Engineering *08/2014 – 08/2016*
University of Engineering and Technology, Taxila, Pakistan

BSc, Telecommunication Engineering *11/2010 – 06/2014*
University of Engineering and Technology, Taxila, Pakistan

Selected Research & Thought Leadership

An AI-Driven Framework for Enhancing Resilience in Propagation Models to Enable Digital Twin 2024
IEEE PIMRC — robust, data-driven modeling and evaluation under limited real data.

Real-World Medication Adherence Monitoring Using Hybrid Neural Networks and Smartwatches 2026
HealthINF (Accepted) — deep learning, multimodal wearable sensing, cross-user evaluation, and real-world reliability.

On Multi-Parameter Optimization and Proactive Reliability in 5G and Beyond Cellular Networks 2025
Sensors — data-driven optimization, reliability analysis, and proactive decision support.

Data-Driven Intelligent Outage Management for High Shadowing Environments in 5G&B Networks 2024
IEEE SmartNets — machine-learning-based outage detection and management for challenging radio environments.

Towards Positioning Error Impact Characterization and Minimization in User-Centric RAN 2022
IEEE WCNC — data-driven characterization, optimization, and performance analysis for user-centric radio access networks.

Toward a Hybrid RF/Optical Lunar Communication System (LunarComm) 2022
IEEE Network — system architecture and design tradeoffs for resilient, high-data-rate hybrid connectivity.

Technical Leadership & Delivery

- Translate ambiguous stakeholder needs into measurable analytics and AI objectives, data-readiness requirements, validation plans, and production-oriented implementation steps.
- Standardize reusable preprocessing, model-evaluation, error-analysis, and experiment workflows to improve reproducibility, reliability, and knowledge transfer across technical work.
- Mentor junior researchers and collaborate with engineering, software, domain, and public-safety stakeholders through technical reviews, experiment planning, documentation, and presentation of results.
- Contribute to thought leadership through peer-reviewed publications, international technical presentations, applied prototypes, and continuous professional learning in GenAI, agentic AI, cloud platforms, and system design.